

# He Min

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## EDUCATIONAL BACKGROUND

**Nanyang Technological University (QS12)**

*Master of Science in Signal Processing and Machine Learning*

**Chongqing University of Technology**

*Bachelor of Science in Computer Science and Technology (Honours)*

- **GPA:** 3.96 / 5.0 (**Top 1%**)

**Singapore**

2026.01-2027.07

**Chongqing, China**

2021.09-2025.06

## INTERNSHIP EXPERIENCE

**Lenovo (Shanghai) Information Technology Co., Ltd. (LR Robotics Algorithm)** 2026.05 ~ 2026.08

**Research Intern in Embodied AI for Robotics**

**Project Overview:** Leveraging the Lenovo Morningstar robot as the platform, a collaborative intelligent agent system has been constructed, comprising the task-planning "mastermind" NavAgent and the dual-system navigator InternVLA-N1-DualVLN. NavAgent utilizes the LangGraph state machine and MCP to orchestrate the ROS2 toolkit, handling global task parsing, memory storage, telemetry self-checks, and anomaly diagnosis. DualVLN manages specific motion generation: System 2 (2 Hz; fine-tuned QwenVL-2.5) predicts 2D pixel sub-goals, while System 1 (30 Hz; NextDiT) outputs high-frequency physical trajectories. SR increased by 10.1%, SPL increased by 6.7%, NE decreased by 17.6%, and StR decreased by 32.5%.

## PROJECT EXPERIENCE

**OV-LidarMap: Open-Set Instance-Level 3D Semantic Mapping with Rotating LiDAR and Panoramic Vision** 2026.01-2026.03

Co-first Author, **IEEE RA-L (SCI Q2 TOP, Under Review)** [[arXiv](#)]

Supervisor: Prof. Lihua Xie

**Project Overview:** To address the issues of lacking spatial reasoning capabilities in Large Language Models (LLMs) and the "context explosion" problem within large-scale 3D environments, an omnidirectional perception-driven embodied AI agent framework was developed using Python, PyTorch, and ROS2. By constructing a dynamic scene graph as the agent's world model and employing a multi-resolution spatial attention mechanism, the framework achieves zero-shot long-range navigation with a Success Rate (SR) of 93.18% and a 69.98% reduction in token costs.

## PUBLICATION

He, M. (2025). A Review of Data Fusion and Deep Learning Models for Multimodal Sentiment Analysis. In Proceedings of E-commerce and Artificial Intelligence (ECAI 2024). [[SciTePress](#)]

## AWARDS & HONORS

**National Scholarship** (2024); First Prize, National College Student Mathematics Competition (2023); Third Prize, National Finals of the Team Programming Ladder Tournament (2023)

## TECHNOLOGY STACK

- **Embodied AI & LLM Agents:** Proficient in designing **state-machine networks** using **LangGraph/LangChain**; deep understanding of the **MCP protocol** with experience in encapsulating **tool ecosystems**; familiar with **Actor-Critic** decision frameworks, **Chain-of-Thought (CoT)** reasoning, and techniques for **LLM prompt compression** and **adaptive resolution**.
- **Robotic Systems & 3D Perception:** Proficient in **ROS 2** communication mechanisms and **Action/Service architectures**; familiar with **SAM2** object segmentation, **Open3D** point cloud processing, and **KD-Tree** spatial search algorithms; experienced in constructing **3D dynamic scene graphs** and designing **hybrid geometric-topological routing**.
- **Systems Engineering & Concurrent Programming:** Proficient in **Python** and **C++**; familiar with **Linux**, **Docker** containerization, and **Git** version control; experienced in encapsulating **low-level control logic** for **multi-threaded applications** and conducting **hardware-software integration and debugging**.